EXPERIMENT NO: - 12

AIM: - To implement various Stack and branch instructions involving insertion and deletion of contents into stack, Conditional and unconditional jump, call and return.

THEORY:

Intel 8085 Instructions: An instruction of a computer is a command given to the computer to perform a specified operation on given data. In microprocessor, the instruction set is the collection of the instructions that the microprocessor is designed to execute.

Branch Control Group: This group contains the instructions for conditional and unconditional jumping, subroutine call and return, and restart.

Unconditional Jump

Instruction set	Explanation	States	Flags	Addressing	Machine cycle
JMP addr(label) [PC] ← Label	Unconditional jump: - jump to the instruction specified by the address	10	None	Immediate	3

Conditional Jump

Instruction set	Explanation	States	Machine cycle
Jump addr (label) [PC] ← Label	Conditional jump: - jump to the instruction specified by the address if the specified condition is fulfilled	10, if true and 7, if not true	3, if true and 2, if not true

Instruction set	Explanation	Status	States	Flags	Addressing	Machine cycle
JZ addr (label) [PC] ←address (label)	Jump, if the result is zero	Jump if Z=1	7/10	None	Immediate	2/3
JNZ addr (label)[PC]	Jump if the result is not zero	Jump if Z=0	7/10	None	Immediate	2/3
Address (label)						
JC addr (label) [PC] ←address (label)	Jump if there is a carry	Jump if CS =1	7/10	None	Immediate	2/3
JNC adder (label) [PC]←address (label)	Jump if there is no carry	Jump if CS =0	7/10	None	Immediate	2/3
JP addr (label) [PC] ←address (label)	Jump if result is plus	Jump if S=0	7/10	None	Immediate	2/3
JM addr (label) [PC] ←address (label)	Jump if result is minus	Jump if S=1	7/10	None	Immediate	2/3
JPE addr(label) [PC]←address (label)	Jump if even parity	The parity status P=1	7/10	None	Immediate	2/3
JPO addr (label) [PC]←address (label)	Jump if odd parity	The parity status P=0	7/10	None	Immediate	2/3

Unconditional CALL

Instruction set	Explanation	States	Flags	Addressing	Machine cycle
CALL addr (label) [SP]-1] ← [PCH] , [[SP-2] ← [PCL], [SP] ← [SP]-2, [PC] ←addr(label)	Unconditional CALL: - Call the subroutine identified by the address	18	None	Immediate /Register	5

Conditional CALL

Instruction set	Explanation	States	Machine cycle
CALL addr (label)[SP]-1] ← [PCH], [[SP-2] ← [PCL], [PC] ← addr (label), [SP]← [SP]-2	Unconditional CALL: Call the subroutine identified by the address if the specified conditional is fulfilled	18, if true and 9, if not true	5, if true and 2, if not true

Instruction set	Explanation	Status	States	Flags	Addressing	Machine cycle
CC addr(label)	Call subroutine if carry status CS=1	CS=1	9/18	None	Immediate /Register	2/5
CNC addr(label)	Call subroutine if carry status CS=0	CS=0	9/18	None	Immediate /Register	2/5
CZ addr(label)	Call subroutine if the result is zero	Zero status Z=1	9/18	None	Immediate /Register	2/5
CNZ addr(label)	Call subroutine if the result is not zero	Zero status Z=0	9/18	None	Immediate /Register	2/5
CP addr (label)	Call subroutine if the result is plus	Sign status S=0	9/18	None	Immediate /Register	2/5
CM addr (label)	Call subroutine if the result is minus	Sign status S= 1	9/18	None	Immediate /Register	2/5

CPE addr (label)	Call subroutine if even parity	Parity Status P=1	9/18	None	Immediate /Register	2/5
CPO addr(label)	Call subroutine if odd parity	Parity Status P= 0	9/18	None	Immediate /Register	2/5

Unconditional Return

Instruction set	Explanation	States	Flags	Addressing	Machine cycle
RET [PCL] ← [[SP]], [PCH] ← [[SP] + 1], [SP] ← [SP] + 2	Unconditional RET: Return from subroutine	10	None	Indirect	3

Conditional Return

Instruction set	Explanation	States	Machine cycle
RET[PCL] ← [[SP]], [PCH] ← [[SP]+ [1], [SP] ← [SP] + 2	Conditional RET: Return from subroutine	12, if true and 6, if not true	3, if true and 1, if not true

Instruction set	Explanation	Status	States	Flags	Addressing	Machine cycle
RC	Return from subroutine if carry status is zero.	CS =1	6/12	None	Register indirect	1/3
RNC	Return from subroutine if carry is not zero	CS = 0	6/12	None	Register indirect	1/3
RZ	Return from subroutine if result is zero.	Zero status Z=1	6/12	None	Register indirect	1/3
RNZ	Return from subroutine if result is not zero.	Zero status Z=0	6/12	None	Register indirect	1/3
RP	Return from subroutine if result is not plus.	Sign Status S= 0	6/12	None	Register indirect	1/3
RM	Return from subroutine if result is not minus.	Sign Status S= 0	6/12	None	Register indirect	1/3
RPE	Return from subroutine if even parity.	Parity Status P= 1	6/12	None	Register indirect	1/3
RPO	Return from subroutine if odd parity.	Parity Status P= 1	6/12	None	Register indirect	1/3